Result & Conclusion

Cases:

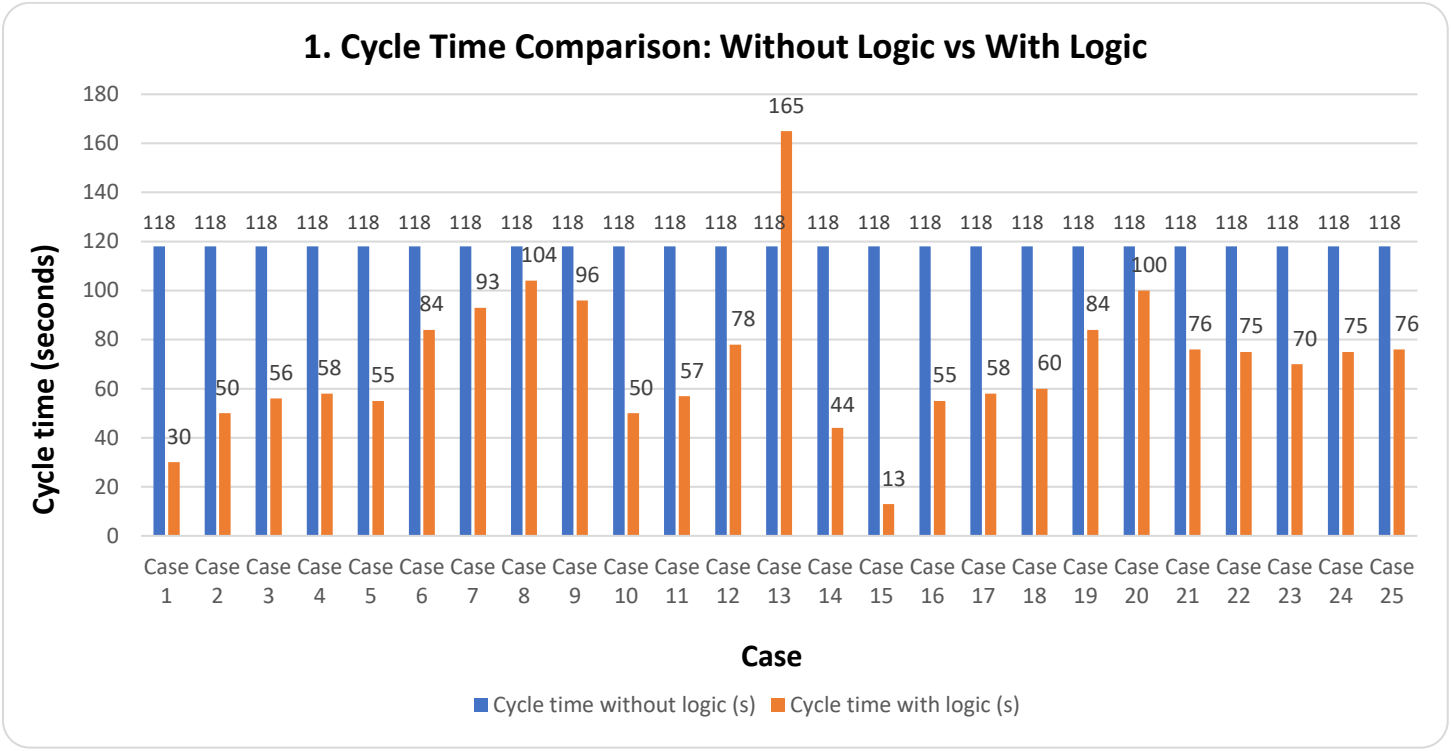
1. Less congestion, No E.V.
2. Less congestion, 1 E.V.
3. Less congestion, 2 E.V.
4. Less congestion, 2 E.V. in same lane
5. More congestion, No E.V.
6. More congestion, 1 E.V. Green Time **> 10 seconds for the previous lane; it will start from the lane next to the previous lane.**
7. More congestion, 1 E.V. Green Time **< 10 seconds for the previous lane.**
8. More congestion, 2 E.V.
9. More congestion, 2 E.V. in same lane
10. Bike count more, without E.V.
11. Bus count more, without E.V.
12. Truck count more, without E.V.
13. E.V. count more.
14. One lane is empty, and the other lanes are congested.
15. Only one lane has vehicles.
16. Less congestion, 1 E.V. Green Time **> 10 seconds for the previous lane; it will start from the lane next to the previous lane.**
17. Less congestion, 1 E.V. Green Time **< 10 seconds for the previous lane.**
18. Bike count more, with 1 E.V.
19. Bus count more, with 1 E.V.
20. Truck count more, with 1 E.V.
21. 2 E. V. at same time (1 Fire truck, 1 ambulance)
22. 2 E. V. at same time (2 ambulance)
23. 2 E. V. at same time (2 Fire truck)
24. 4 E. V. at same time (2 ambulance, 2 fire truck)
25. 4 E. V. at same time (4 ambulance)

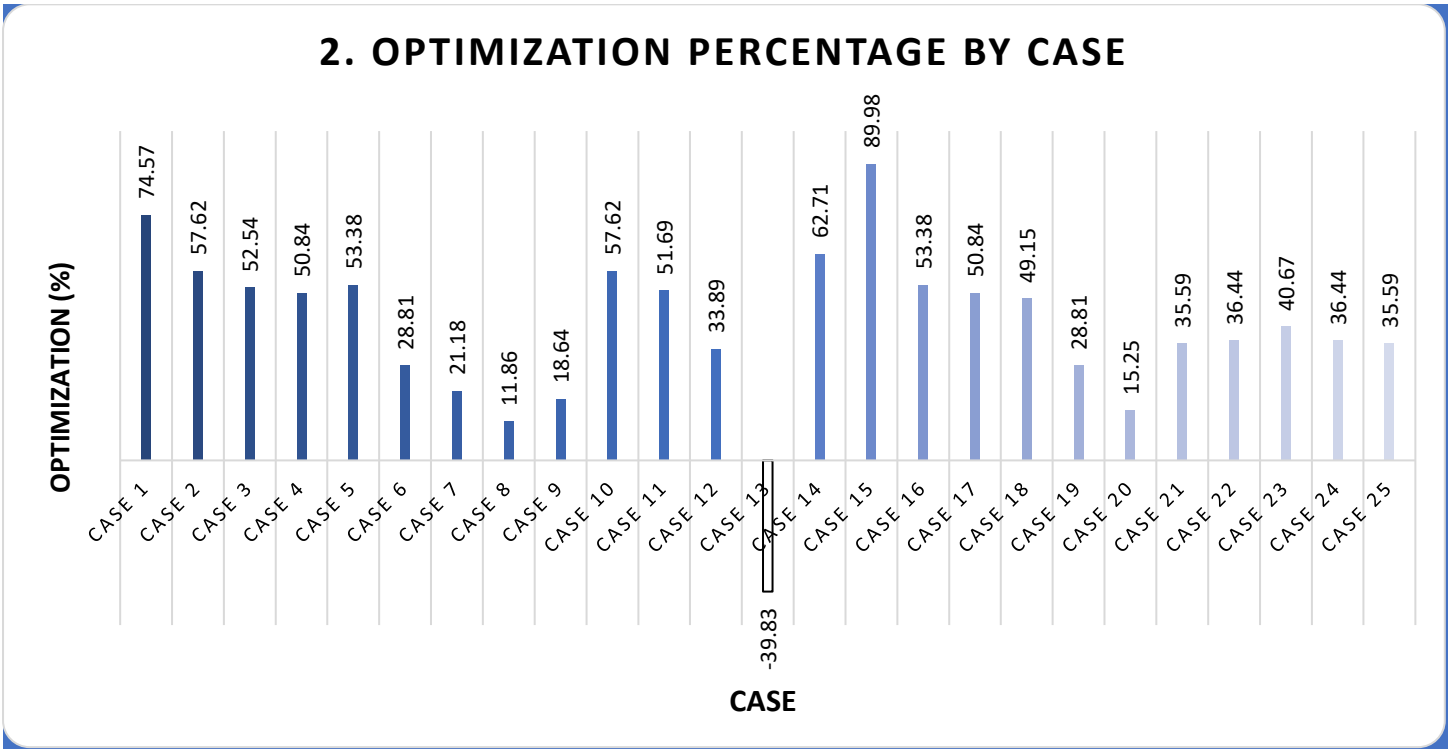
Observation Table :

Result	Time without logic	Time with logic	optimization (%)
Case - 1	118	30	74.57
Case - 2	118	50	57.62
Case - 3	118	56	52.54
Case - 4	118	58	50.84
Case - 5	118	55	53.38
Case - 6	118	84	28.81
Case - 7	118	93	21.18
Case - 8	118	104	11.86
Case - 9	118	96	18.64
Case - 10	118	50	57.62
Case - 11	118	57	51.69
Case - 12	118	78	33.89
Case - 13	118	165	-39.83
Case - 14	118	44	62.71
Case - 15	118	13	89.98
Case - 16	118	55	53.38
Case - 17	118	58	50.84
Case - 18	118	60	49.15
Case - 19	118	84	28.81
Case - 20	118	100	15.25
Case - 21	118	76	35.59
Case – 22	118	75	36.44
Case – 23	118	70	40.67
Case – 24	118	75	36.44
Case – 25	118	76	35.59
	Avg = 118	Avg = 71.48	Avg = 40.27

$$\textbf{Optimization (\%)} = \frac{\textbf{Time without logic} - \textbf{Time with logic}}{\textbf{Time without logic}} \times 100$$

Graphs :


1. Cycle Time Comparison: Without Logic vs With Logic
Cycle time (seconds)
180
160
140
120
100
80
60
40
20
0
118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118 118
30 50 56 58 55 84 93 104 96 50 57 78 165 44 13 55 58 60 84 100 76 75 70 75 76
Case 1 Case 2 Case 3 Case 4 Case 5 Case 6 Case 7 Case 8 Case 9 Case 10 Case 11 Case 12 Case 13 Case 14 Case 15 Case 16 Case 17 Case 18 Case 19 Case 20 Case 21 Case 22 Case 23 Case 24 Case 25
Case
Cycle time without logic (s)
Cycle time with logic (s)


2. OPTIMIZATION PERCENTAGE BY CASE
OPTIMIZATION (%)
74.57 57.62 52.54 50.84 53.38 28.81 21.18 11.86 18.64 57.62 51.69 33.89 -39.83 62.71 89.98 53.38 50.84 49.15 28.81 15.25 35.59 36.44 40.67 36.44 35.59
CASE 1 CASE 2 CASE 3 CASE 4 CASE 5 CASE 6 CASE 7 CASE 8 CASE 9 CASE 10 CASE 11 CASE 12 CASE 13 CASE 14 CASE 15 CASE 16 CASE 17 CASE 18 CASE 19 CASE 20 CASE 21 CASE 22 CASE 23 CASE 24 CASE 25
CASE

3. EMERGENCY VEHICLE HANDLING: CYCLE TIME BY SCENARIO

